SPE-230168-MS

Cavitation-Assisted Intensified Oxidation Solutions to Real Industrial Wastewater Streams in KSA

Khalid Alkhayat, Ümit Görçüm, Shekhar R. Kulkarni, Varaha P. Sarvothaman, Deoras M. Prabhudharwadkar, and William L. Roberts, Clean Energy Research Platform, King Abdullah University of Science and Technology, Saudi Arabia

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Human life relies on energy and water resources for bodily functions, food cultivation, and daily materials. Agricultural and industrial processes introduce toxic chemicals into wastewater, with industrial wastewater accounting for 15% of global water use. These wastewaters contain engineered substances that cannot be discharged into natural habitats due to high organic and solid content. When introduced to freshwater, they cause chronic issues for humans and aquatic life[1,2]. The secondary stage of wastewater treatment deals with broad oxidation processes. Conventional methods, like activated sludge and trickling filters, have drawbacks such as long treatment times and large land requirements, highlighting the need for more efficient methods[3,4]. Advanced oxidation processes (AOPs) are those which can generate reactive radicals which help break down chemical moieties in water, otherwise found challenging for conventional oxidation processes. AOPs can be classified into those which can be initiated by adding chemicals such as hydrogen peroxide, ozone or chlorine containing chemicals, electrochemical based ones and those which are based on catalyst's (such as Fenton's oxidation and photocatalysis[5]).

Fenton's treatment is an aggressive chemical treatment, which is highly capable of oxidizing large pollutant concentrations[6]. Fenton's chemistry has been used as a reliable treatment technique for different effluents and is industrially implemented. The rapid oxidation of iron (II) to iron (III), and its regeneration, to produce a high and steady quantity is preferred for extremely high organic content waters. However, the sludge formed and the rapid increase in temperatures are disadvantages of this technique when applied to real industrial environments. The reactions of this chemistry are listed in reactions (R1 to R4) below.

Introduction

Human life relies on energy and water resources for bodily functions, food cultivation, and daily materials. Agricultural and industrial processes introduce toxic chemicals into wastewater, with industrial wastewater accounting for 15% of global water use. These wastewaters contain engineered substances that cannot be discharged into natural habitats due to high organic and solid content. When introduced to freshwater, they cause chronic issues for humans and aquatic life[1,2]. The secondary stage of wastewater treatment deals with broad oxidation processes. Conventional methods, like activated sludge and trickling filters, have drawbacks such as long treatment times and large land requirements, highlighting the need for more efficient methods[3, 4]. Advanced oxidation processes (AOPs) are those which can generate reactive radicals which help break down chemical moieties in water, otherwise found challenging for conventional oxidation processes. AOPs can be classified into those which can be initiated by adding chemicals such as hydrogen peroxide, ozone or chlorine containing chemicals, electrochemical based ones and those which are based on catalyst's (such as Fenton's oxidation and photocatalysis[5]).

Fenton's treatment is an aggressive chemical treatment, which is highly capable of oxidizing large pollutant concentrations[6]. Fenton's chemistry has been used as a reliable treatment technique for different effluents and is industrially implemented. The rapid oxidation of iron (II) to iron (III), and its regeneration, to produce a high and steady quantity is preferred for extremely high organic content waters. However, the sludge formed and the rapid increase in temperatures are disadvantages of this technique when applied to real industrial environments. The reactions of this chemistry are listed in reactions (R1 to R4) below.

$$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^- + OH\cdot \qquad \text{R 1}$$

$$Fe^{3+} + H_2O_2 \rightarrow Fe - OOH^{2+} + H^+ \qquad \text{R 2}$$

$$Fe - OOH^{2+} \rightarrow Fe^{2+} + HO_2^{\cdot} \qquad \text{R 3}$$

$$H_2O_2 \rightarrow 2\,OH\cdot \qquad \text{R 4}$$

Among other catalyst based AOPs, semi-conductor photocatalysis, a process using metal oxides as photocatalyst, is a promising strategy without external chemicals. When these metal oxides, such as ZnO and TiO_2, illuminated with a light of suitable wavelength, highly oxidizing species are produced[7]. This process driven by light is a powerful tool to produce oxidants in the liquid phase and drive oxidation of organic pollutants in wastewater. The reactions of this chemistry technique are listed in reactions (R5 to R9) below. The advantages of this technique include low operating costs, reducing the need for excessive external oxidant dosing and does not suffer from sludge formation unlike the Fenton's process. Particle agglomeration and limitations due to mass transfer exist, which can be overcome by suitable techniques such as incorporating with cavitation. The photocatalytic technique has gained prominence with a great emphasis on material development[8], application to different waste streams[9], coupling with cavitation[10] and so on. Despite the promise of the photocatalysis process on a lab-scale to treat various effluent streams, mass transfer limitations stunt its scalability. The coupling of cavitation with photocatalysis helps overcome mass transfer limitations and helps in intermittently 'cleansing' the catalyst surface and promoting the degradation of pollutants.

$$Photocatalyst + h\vartheta \rightarrow h_{vb}^+ + e_{cb}^- \qquad \text{R 5}$$

$$h_{vb}^+ + H_2O \rightarrow OH^{\cdot} + \mathrm{H}^+ \qquad \text{R 6}$$

$$h_{vb}^+ + OH^- \rightarrow OH^{\cdot}_{ad} \qquad \text{R 7}$$

$$h_{vb}^+ + R_{ads} \rightarrow R^+ \qquad \text{R 8}$$

$$e_{cb}^- + O_2 \rightarrow O_2^- \qquad \text{R 9}$$

Cavitation, the formation of microbubbles as a result of dynamic pressure changes in the liquid has been applied to intensify water treatment by both the Fenton's[4, 11, 12] and photocatalytic techniques[12, 13, 14]. While the application of cavitation-Fenton's treatment is beyond doubt by the cited literature in driving chemistry for very large organic content, the coupling of cavitation-photocatalysis has reported the handling of volumes ranging from 5 to 800 L batches, however, the concentrations considered do not exceed 60 parts per million (60 milligrams per Liter), which are lesser than the indicative chemical oxygen demand (COD) of industrial effluents (up to 140,000 ppm)[2, 15]. The upper limit of COD here is indicated to be much higher than usual, a range of COD in the range of 2,000 to 5,000 ppm can be considered a typical range. It is beyond doubt that photocatalysis and photocatalysis coupled with cavitation, can provide high removal rates (>90%) in the range up to 100 ppm of pollutant[13]. It is important to evaluate the feasibility of photocatalysis for higher pollutant concentrations, and the need for supplementary oxidants and their interaction while handling higher pollutant concentrations. By doing so, the applicability of photocatalytic based oxidation can be examined and potentially be deployed. Figure 1 shows the applicability of diverse chemistries driven by cavitation to different feeds.

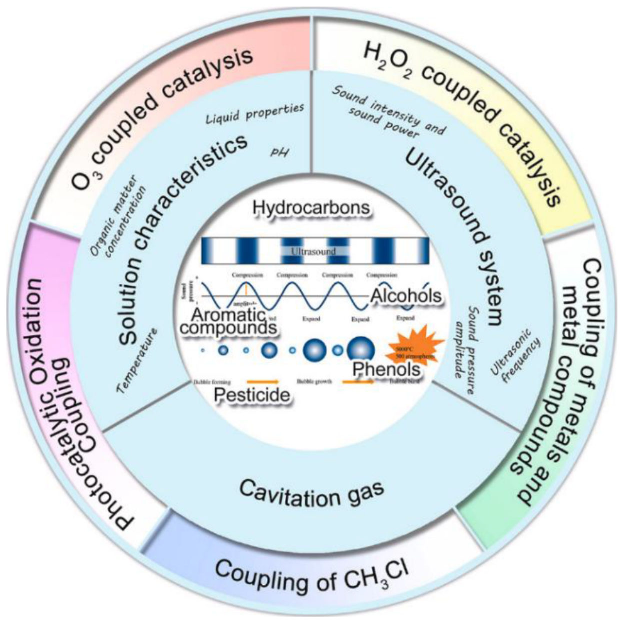

Figure 1—Schematic on the application of acoustic cavitation to water treatment [Song et al. (2024)][16].

The overarching aim of AOPs is to reduce the resources specifically external chemical dosing to these water treatment processes. While the goal is to reduce the load of external chemical oxidants, the exact scheme of implementation of AOPs varies and is dependent on the process goal: AOPs can be implemented after primary, secondary and desalination steps, and sometimes to reduce the toxicity of a stream prior to biological treatment[17]. Cavitation has been typically applied to dilute contaminated streams, and not very contaminated streams as explored in this work. We aim to use cavitation as a tool to intensify oxidation, by exploring the applicability of Fenton's and photocatalysis to 3 different industrial streams and propose a suitable oxidative pathway. In addition to this, the quantification of cavitation-based water treatment will be calculated, to highlight the beneficial effect of cavitation.

Materials and methodology

Zinc oxide (ZnO, CAS number: 1314–13-2), iron oxide ($FeSO_4$.7 H_2O, CAS number: 7782-63-0) and hydrogen peroxide (H_2O_2, 30 and 40% strengths), were procured from Sigma Aldrich. UV-LED light strips (Model: SMD3528, Wavelength: 370 nm, Power specifications: 48 W) were procured from Amazon UK. Visible light strips (Model: Mextronic LED Strip 3528, Wavelength: 465-470 nm, Power specifications: 24 W) were procured from Amazon DE. The industrial effluents were collected from various locations in the Kingdom of Saudi Arabia (KSA).

Experiments on the beaker-scale were carried out with a total volume of 200-mL (Figure 2 a and 2b). Acoustic cavitation was generated using an ultrasonic probe (Hielscher UP400St) was inserted in the beaker 30 mm in the solution for the beaker-scale experiments. The industrial effluents were sampled from different chemical and allied industries, the organic and solids content were quantified as total organic carbon (TOC), and total dissolved solids (TDS) were measured. These values are tabulated in Table 1. No further analysis of the source of the effluent, specific organic compounds or metals as a consequence of TDS are reported due to confidentiality.

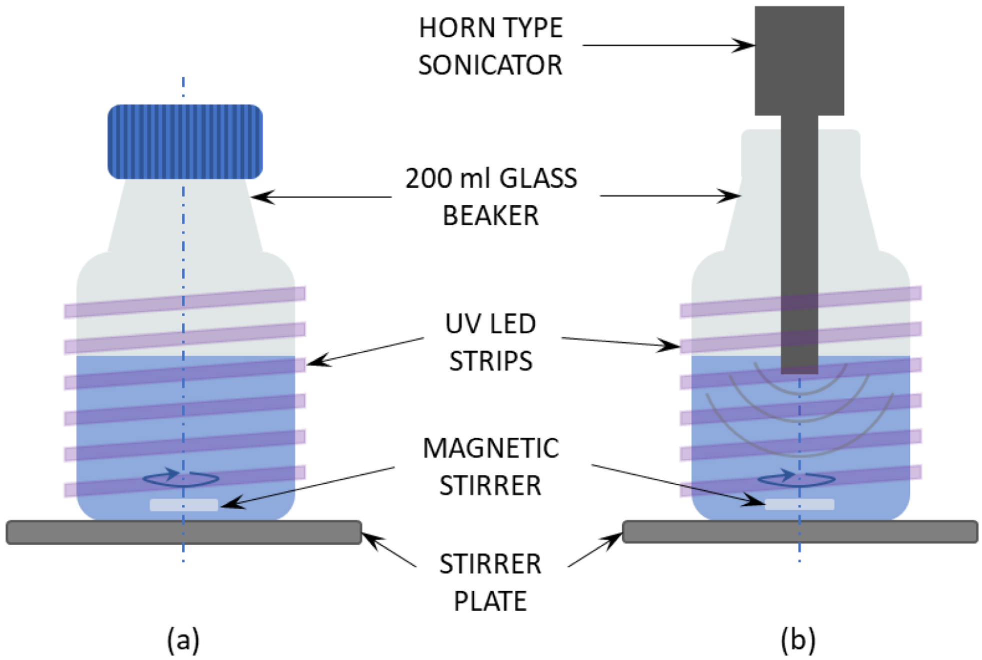

Figure 2—Schematic of experimental setup for experimentation on the beaker scale – photocatalytic based oxidation (left) and cavitation assisted photocatalytic oxidation (right).

Table 1—Characterization of industrial effluents.

Stream #	TOC content (ppm)	TDS (ppm)	Initial pH
1	25,000 – 55,000	15,000 – 35,000	9 – 12.5
2	8,500 – 15,500	75,000 – 105,000	3.5 – 5
3	2,500 – 5,500	8,000 – 26,000	5 – 7.5

Results

Screening of different oxidation strategies – suitability of Fenton's treatment

The treatment of effluents with hydrogen peroxide (H_2O_2) is known to perform better under acidic conditions. However, too much addition of acid hampers the reactivity of H_2O_2. Direct addition of H_2O_2 was attempted on stream 1, however, this resulted in intense frothing (shown in Figure 3). On a large scale of treatment, defoaming agents and adjusting aeration is possible. However, on a small scale, it was acidified using acid and then handled. Moreover, acidic conditions favored oxidant-based treatment. To progress to Fenton's treatment, $FeSO_4$. 7 H_2O was added to a mixture of effluent stream 1 and H_2O_2, however, upon iron salt addition a vigorous frothing could not be avoided.

Stream 3 exhibited a similar behavior to Fenton's treatment, and hence iron salt addition-based oxidation was ruled out for streams 1 and 3.

Figure 3—Action of hydrogen peroxide on industrial stream 1 at unaltered pH.

Treatment with 30% hydrogen peroxide and short processing times

Experiments were carried out with streams 2 and 3 on a 50-mL beaker scale, to avoid excessive use of the sampled industrial effluent. These were carried out for 10 minutes of stirring or sonication, with 30% H_2O_2, where the experiments comprised only H_2O_2, Fenton's treatment and photocatalytic treatment to screen promising oxidative methods and build on them in further work. 20 experiments were conducted in this exercise. Of which the stand-out experiment was subjecting stream 3 to photocatalytic treatment, and it was observed that a 'clear' water, this is possible with leaving the oxidation on for 12 hours, and a 30% oxidative removal without cavitation was observed here. The destruction of color in stream 3 is possibly due to the breakage of conjugated compounds or specifically azo bonds, which might have been destroyed to form carboxylic acids or a shorter chain than the parent pollutants. The 70% remnant TOC is the challenge. Some other observations from this 20-experiment exercise were that Fenton's treatment is highly exothermic with stream 3 elevating the temperature to 85°C unless iron addition is optimized.

The key takeaway from this brief exercise was to limit iron sulfate addition to avoid excessive heat generation. And it must be noted that the experiments performed under sonication do not allow an isothermal operation and heat up the solution. For instance, a 5-minute operation of a 50-mL volume at 100% duty cycle heats ambient temperature effluent to 50 °C, which is very harmful for preserving oxidants such as H_2O_2 in the liquid phase and thus skewing the TOC values.

Parametric studies with 40% hydrogen peroxide and longer processing times

Building on the promise from the earlier exercise of 20 experiments (in Section 3.2), it was decided to investigate streams 2 and 3 with a systematic dosing of 40% H_2O_2 and for longer treatment times. This meant that the experiments without cavitation comprised 180 minutes of operation with samples gathered at 0, 5, 15, 30, 60, 120 and 180 minutes. And the experiment with cavitation comprised a 45-minute operation with samples gathered at 0, 3, 15, 21 and 45 minutes. The experiments with cavitation were cooled from 50/55 °C back to room temperature with the help of ice. 40% H_2O_2 was employed to limit the addition of a larger volume of chemical diluent and utilize the peroxide better.

For stream 2, 4 experiments were performed of H_2O_2/TOC ranged from 0.75 to 4.5, and the Fenton's and photocatalytic treatment were performed at a catalyst loading of 2.5 g/L at a H_2O_2/TOC = 0.75 and 3.2 respectively. Thus, these were 6 experiments of chemical variants, and this was multiplied by 2 to get the effect of without and with cavitation. A total of 12 experiments were conducted. Similarly for stream 3, 4 experiments were performed of H_2O_2/TOC ranged from 0.75 to 7.5, and the Fenton's and photocatalytic treatment were performed at a catalyst loading of 2.5 g/L at a H_2O_2/TOC = 0.75 and 4.8 respectively. Thus, these were 6 experiments of chemical variants, and this was multiplied by 2 to get the effect of without and with cavitation. A total of 12 experiments were conducted.

For the experiments of stream 2 (no cavitation), the results showed a 4-9% TOC removal for the experiments which led to the conclusion that stream 2 can be degraded with H_2O_2. However, the experiments ranging from H_2O_2/TOC ranged from 0.75 to 4.5 did not show a monotonic increase in the degradation with increase in H_2O_2 loading. It is possible that the exothermic reaction between peroxide and the effluent promoted volatility, hence the stripped peroxide ends up concentrating the TOC at very high loadings of oxidant. In the case of experiments with Fenton's and photocatalytic treatment showed a 6 or 10% increase in TOC which plainly indicates that the dosed peroxide was no longer present. *For the experiments of stream 2 (with cavitation)*, the experiments consistently showed a 11 – 38% increase in TOC, which meant that the 40% peroxide certainly had an exothermic effect coupled with the heating introduced by cavitation, which stripped excessive peroxide within a 45-minute period. Here only the final samples were analyzed which were the 180th minute sample for no cavitation and 45th minute samples for cavitation-based treatment.

For the experiments of stream 3 (no cavitation), unlike the experiments of stream 2 which showed removal of TOC at different doses of peroxide, the experiments here showed a 4 – 15% increase in TOC for all experiments (H_2O_2, Fenton's and photocatalysis), which meant that the exothermicity or volatility of peroxide was present in the case of stream 3 water too. *For the experiments of stream 3 (with cavitation)*, the experiments in this category showed a 27-50% increase in TOC, which meant that the coupled effect of exothermicity and volatility played a major role.

Overall, the testing with 40% H_2O_2, although which was planned as 24 experiments with long durations (45 minutes – with cavitation and 180 minutes – without cavitation), and with the overarching aim of gathering systematic data proved to be unsatisfactory. The findings overall indicate that the addition of 40% H_2O_2 at the commencement of the experiment is a bad strategy due to improper utilization of the oxidant, thus, it will help to add less concentrated doses in a stepwise manner to utilize the oxidant better. Hence the rest of the testing was decided to be with 30% H_2O_2 for demonstrating a TOC removal.

Optimized recipe and quantification of cavitation-based treatment

Experiments were conducted at acidic pH conditions, with photocatalytic treatment carried out for streams 1 and 3, and Fenton's treatment carried out for stream 2. The chemical ratios are as tabulated in Table 2.

Table 2—Oxidant chemicals dosed.

Stream #	pH adjustment	30% H_2O_2 added (% of reactor volume)	Catalyst added (g/L)
1	To 6.5	6.25	1.25 – TiO_2
2	None	12.75	1.875 – $FeSO_4.7H_2O$
3	None	9.5	2.5 – TiO_2

The experiments mentioned in Table 2 were carried out for a uniform time of processing (time = 9-minutes), these revealed at least a 4.67x improvement in the utilization of oxidants by cavitation when compared to no-cavitation conditions, which was for the case of stream 2. In the cases of streams 1 and 3, these were a 7 and 6-fold improvement by the use of cavitation (Figure 4). The grade of peroxide used was 30% peroxide and the treatment time was limited to 9-minutes in this case which helped in quantifying a finite degradation. Whereas the use 40% peroxide (Section 3.3) resulted in more exothermic reactions, causing more volatility and possibly the evaporation of peroxide-water mixture, which resulted in a net-TOC increase.

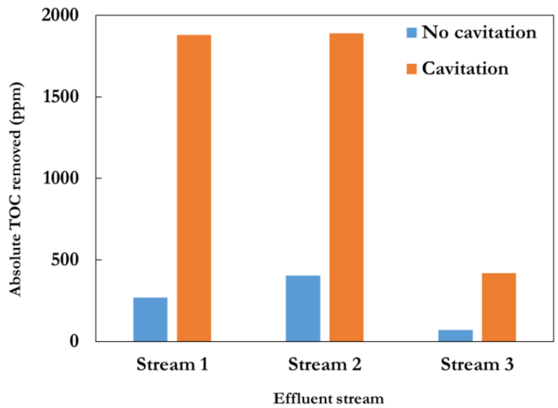

Figure 4—Influence of cavitation on TOC removal for the respective streams for optimized oxidation route.

Costing of a photocatalytic based advanced oxidation operation

In this work the use of a photocatalytic approach worked with 2 of 3 effluents, and hence a costing is performed for the photocatalytic route of processing. Often, the cost of an emerging technology such as photocatalysis is compared to conventional processing. However, the cost of conventional technologies is sensitive to the feed and materials used for treatment. Hence it is hard to establish a comparison of existing techniques for the effluents used in this work. Despite this, a few assumptions are made to estimate the costs of a 1000 Tons per day (TPD) of effluent 3, to provide a representative costing, identify the split between capital expenditure (CAPEX) and operating expenditure (OPEX), which will provide directions for future work.

Considering 40 acoustic cavitators to process 1000 TPD priced at $20,000 each, 40 UV lighting arrangements to power photocatalysis in these cavitation chambers priced at $100 each, and 35 centrifuges to remove photocatalyst from the processed slurry priced at $12,000 each and power requirements, the overall CAPEX results in $2.12 M. Considering 20 years of operation, and a linear depreciation, the normalized CAPEX is $0.29/ton, and the rest of the costs split between oxidizer, utilities, and catalyst are presented in Figure 5.

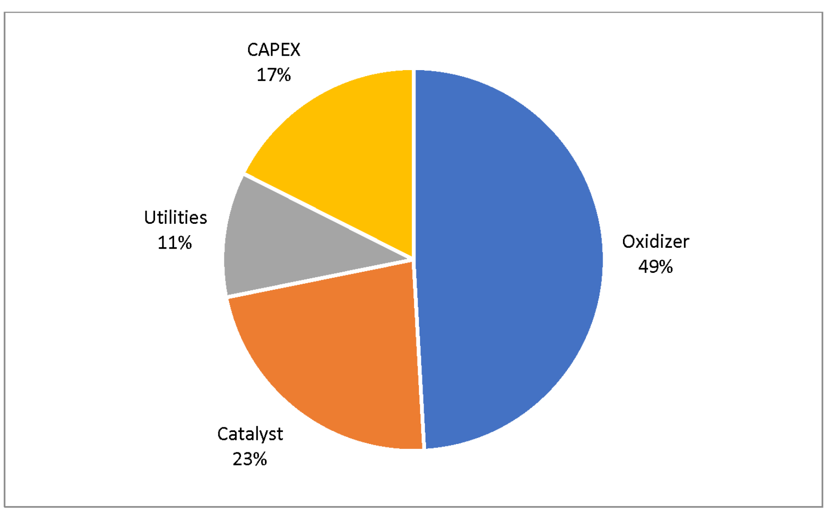

Figure 5—Cost split between different components of a 1000 Tons per day photocatalytic treatment, for 90% conversion on an effluent with initial TOC content of 2500 ppm.

Table 3—Calculation of treatment cost for effluent 3.

Scale			
Treatment capacity (TPD)	1000	**Costing**	
Conversion		Pure H_2O_2 required ($/ton)	1200
TOC_0 (ppm)	2500	ZnO powder ($/ton)	3000
Conversion (%)	90		
Final TOC (ppm)	250	Pure H_2O_2 cost ($)	810
		ZnO cost ($)	375
Catalyst			
Process required (g/L)	2.5	Total OPEX ($)	1185
% Recovery	95	OPEX ($/ton)	1.185
Consumption (%)	5		
Consumption (g/L)	0.125		
Oxidant			
30% H_2O_2 required (mg/L)	2250		
Pure H_2O_2 required (mg/L)	675		
Pure H_2O_2 required (g/L)	0.675		
H_2O_2/TOC (-)	0.3		

From Figure 5 it is evident the oxidant is responsible for half of the costs in treating effluents, and reducing this cost can help reduce the overall expenses toward photocatalytically mediated advanced oxidation processes. With approaches such as doping of photocatalysts and tailoring them to specific wastewater characteristics[18], it will be possible to reduce the cost of external oxidants, and ultimately aid the reduction of treatment costs.

Conclusions

Industrial water treatment enables water recycling and reuse, which can help the overall sustainability of industrial processing. At present, the chemicals being dosed for the secondary stage of water treatment increase operating costs, and this can be brought down by process intensification methods such as cavitation. In this work, acoustic cavitation was applied to chemically mediated oxidative degradation of industrial effluents. And this resulted in a 5-7 time increase in total organic carbon removal for 3 different industrial streams obtained within KSA. These results show the promise of applying cavitation, shortening treatment time for TOC removal, because of more efficient utilization of dosed chemicals. The present work developed a recipe and had a very limited demonstration of the application of cavitation to the effluent streams and needs to be expanded in terms of spanning multiple parameters while simultaneously applying cavitation. The future work can include spanning various relevant parameters such as pH, oxidant loading, catalyst loading, and acoustic cavitation parameters. Overall, the application of cavitation, a process intensification tool, is possible for accelerating oxidative degradation of industrial effluents, and this work is one of the first to apply it to multiple streams and optimize a recipe for it. Furthermore, a simple costing was attempted for one of the effluent's in this work. The costing indicated a 50% expense towards oxidants in the photocatalytically mediated advanced oxidation, hence efforts to improve photocatalytic performance will be key in reducing treatment costs and scoping industrial applications.

References

1. Loeb, S. K.; Alvarez, P.J.J.; Brame, J.A.; Cates, E.L.; Choi, W.; Crittenden, J.; Dionysiou, D.D.; Li, Q.; Li-Puma, G.; Quan, X.; et al The Technology Horizon for Photocatalytic Water Treatment: Sunrise or Sunset? *Environmental Science & Technology* 2019, **53** (6), 2937–2947. DOI: 10.1021/acs.est.8b05041. Verma, S.; Kumar, P.; Srivastava, V. C.; Štangar, U. L. Application of advanced oxidation processes (AOPs) for the treatment of petrochemical industry wastewater. Advanced Industrial Wastewater Treatment and Reclamation of Water: Comparative Study of Water Pollution Index during Pre-industrial, Industrial Period and Prospect of Wastewater Treatment for Water Resource Conservation 2022, 103-128.
2. Sathya, K.; Nagarajan, K.; Carlin Geor Malar, G.; Rajalakshmi, S.; A comprehensive review on comparison among effluent treatment methods and modern methods of treatment of industrial wastewater effluent from different sources. *Applied Water Science* 2022, **12** (4), 70.
3. Parsons, S. *Advanced oxidation processes for water and wastewater treatment*; IWA publishing, 2004. Oller, I.; Malato, S.; Sánchez-Pérez, J. Combination of advanced oxidation processes and biological treatments for wastewater decontamination—a review. Science of the total environment 2011, 409 (20), 4141-4166. Elmobarak, W. F.; Hameed, B. H.; Almomani, F.; Abdullah, A. Z. A review on the treatment of petroleum refinery wastewater using advanced oxidation processes. Catalysts 2021, 11 (7), 782.
4. Eren, Z. Ultrasound as a basic and auxiliary process for dye remediation: A review. *Journal of Environmental Management* 2012, **104**, 127–141. DOI: https://doi.org/10.1016/j.jenvman.2012.03.028.
5. Garrido-Cardenas, J. A.; Esteban-García, B.; Agüera, A.; Sánchez-Pérez, J.A.; Manzano-Agugliaro, F. Wastewater treatment by advanced oxidation process and their worldwide research trends. *International Journal of Environmental Research and Public Health* 2020, **17** (1), 170.
6. Cai, Q. Q.; Jothinathan, L.; Deng, S.H.; Ong, S.L.; Ng, H.Y.; Hu, J.Y. Fenton-and ozone-based AOP processes for industrial effluent treatment. In *Advanced oxidation processes for effluent treatment plants*, Elsevier, 2021; pp 199–254.
7. Lee, S.-Y.; Park, S.-J. TiO2 photocatalyst for water treatment applications. *Journal of industrial and engineering chemistry* 2013, **19** (6), 1761–1769.
8. Akpan, U. G.; Hameed, B.H. Parameters affecting the photocatalytic degradation of dyes using TiO2-based photocatalysts: a review. *Journal of hazardous materials* 2009, **170** (2-3), 520–529. Rajoriya, S.; Bargole, S.; George, S.; Saharan, V. K.; Gogate, P. R.; Pandit, A. B. Synthesis and characterization of samarium and nitrogen doped TiO2 photocatalysts for photo-degradation of 4-acetamidophenol in combination with hydrodynamic and acoustic cavitation. Separation and purification technology 2019, 209, 254-269. Chen, D.; Cheng, Y.; Zhou, N.; Chen, P.; Wang, Y.; Li, K.; Huo, S.; Cheng, P.; Peng, P.; Zhang, R. Photocatalytic degradation of organic pollutants using TiO2-based photocatalysts: A review. journal of cleaner production 2020, 268, 121725.
9. Chemseddine, A.; Boehm, H.P. A study of the primary step in the photochemical degradation of acetic acid and chloroacetic acids on a TiO2 photocatalyst. *Journal of molecular catalysis* 1990, **60** (3), 295–311.

10. Gogate, P. R.; Pandit, A.B. Sonophotocatalytic reactors for wastewater treatment: a critical review. *AIChE journal* 2004, **50** (5), 1051–1079.
11. Wu, Z.; Yuste-Córdoba, F.J.; Cintas, P.; Wu, Z.; Boffa, L.; Mantegna, S.; Cravotto, G. Effects of ultrasonic and hydrodynamic cavitation on the treatment of cork wastewater by flocculation and Fenton processes. *Ultrasonics sonochemistry* 2018, **40**, 3–8. Kumar, M. S.; Sonawane, S. H.; Bhanvase, B. A.; Bethi, B. Treatment of ternary dye wastewater by hydrodynamic cavitation combined with other advanced oxidation processes (AOP's). Journal of Water Process Engineering 2018, 23, 250-256. DOI: https://doi.org/10.1016/j.jwpe.2018.04.004. Cai, M.; Su, J.; Zhu, Y.; Wei, X.; Jin, M.; Zhang, H.; Dong, C.; Wei, Z. Decolorization of azo dyes Orange G using hydrodynamic cavitation coupled with heterogeneous Fenton process. Ultrasonics sonochemistry 2016, 28, 302-310. Angaji, M. T.; Ghiaee, R. Decontamination of unsymmetrical dimethylhydrazine waste water by hydrodynamic cavitation-induced advanced Fenton process. Ultrasonics sonochemistry 2015, 23, 257-265. Gogate, P. R.; Patil, P. N. Combined treatment technology based on synergism between hydrodynamic cavitation and advanced oxidation processes. Ultrasonics Sonochemistry 2015, 25, 60-69. DOI: https://doi.org/10.1016/j.ultsonch.2014.08.016. Bhasarkar, J. B.;; Chakma, S.;; Moholkar, V. S.;. Mechanistic features of oxidative desulfurization using sono-fenton-peracetic acid (ultrasound/ Fe2+-CH3COOH-H 2O2) system. American Chemical Society: 2013; Vol. 52, pp 9038-9047. Pradhan, A. A.; Gogate, P. R. Removal of p-nitrophenol using hydrodynamic cavitation and Fenton chemistry at pilot scale operation. Chemical Engineering Journal 2010, 156 (1), 77-82. Chakinala, A. G.; Gogate, P. R.; Burgess, A. E.; Bremner, D. H. Industrial wastewater treatment using hydrodynamic cavitation and heterogeneous advanced Fenton processing. Chemical Engineering Journal 2009, 152 (2-3), 498-502. Chakinala, A. G.; Gogate, P. R.; Burgess, A. E.; Bremner, D. H. Treatment of industrial wastewater effluents using hydrodynamic cavitation and the advanced Fenton process. Ultrasonics sonochemistry 2008, 15 (1), 49-54.
12. Patil, P. N.; Gogate, P.R. Degradation of dichlorvos using hybrid advanced oxidation processes based on ultrasound. *Journal of Water Process Engineering* 2015, **8**, e58–e65. DOI: https://doi.org/10.1016/j.jwpe.2014.10.012.
13. De-Nasri, S. J.; Sarvothaman, V.P.; Kelly, C.; Nagarajan, S.; Manesiotis, P.; Robertson, P.K.J.; Ranade, V.V. Hydrodynamic cavitation and photocatalysis for effluent treatment: Key operating parameters and synergistic effects. *Chemical Engineering Journal* 2023, 145241. Wang, X.; Jia, J.; Wang, Y. Combination of photocatalysis with hydrodynamic cavitation for degradation of tetracycline. Chemical Engineering Journal 2017, 315, 274-282. Çalışkan, Y.; Yatmaz, H. C.; Bektaş, N. Photocatalytic oxidation of high concentrated dye solutions enhanced by hydrodynamic cavitation in a pilot reactor. Process Safety and Environmental Protection 2017, 111, 428-438. DOI: https://doi.org/10.1016/j.psep.2017.08.003.
14. De-Nasri, S. J.; Nagarajan, S.; Robertson, P.K.J.; Ranade, V.V.; *Quantification of hydroxyl radicals in photocatalysis and acoustic cavitation: Utility of coumarin as a chemical probe*. Elsevier: 2021; Vol. **420**, p 127560. De-Nasri, S. J.; Nagarajan, S.; Robertson, P. K. J.; Ranade, V. V. Quantification of hydroxyl radicals in photocatalysis and acoustic cavitation: Utility of coumarin as a chemical probe. Chemical Engineering Journal 2020, 127560. Bagal, M. V.; Gogate, P. R. Degradation of diclofenac sodium using combined processes based on hydrodynamic cavitation and heterogeneous photocatalysis. Ultrasonics sonochemistry 2014, 21 (3), 1035-1043. Wang, X.; Jia, J.; Wang, Y. Degradation of CI Reactive Red 2 through photocatalysis coupled with water jet cavitation. Journal of Hazardous Materials 2011, 185 (1), 315-321.

15. Bhandari, V. M.; Sorokhaibam, L.G.; Ranade, V.V. Industrial wastewater treatment for fertilizer industry—A case study. *Desalination and Water Treatment* 2016, **57** (57), 27934–27944. Iloms, E.; Ololade, O. O.; Ogola, H. J. O.; Selvarajan, R. Investigating industrial effluent impact on municipal wastewater treatment plant in Vaal, South Africa. International journal of environmental research and public health 2020, 17 (3), 1096.
16. Song, K.; Liu, Y.; Umar, A.; Ma, H.; Wang, H. Ultrasonic cavitation: Tackling organic pollutants in wastewater. *Chemosphere* 2024, **350**, 141024.
17. Coha, M.; Farinelli, G.; Tiraferri, A.; Minella, M.; Vione, D. Advanced oxidation processes in the removal of organic substances from produced water: Potential, configurations, and research needs. *Chemical Engineering Journal* 2021, **414**, 128668.
18. Yang, G.; Chen, H.; Qin, H.; Zhang, X.; Feng, Y. Effect of nitrogen doping on the catalytic activity of activated carbon and distribution of oxidation products in catalytic wet oxidation of phenol. *The Canadian Journal of Chemical Engineering* 2017, **95** (8), 1518–1525.